

1. Model Training Overview

The main goal of model training is to **learn patterns from data** that can predict the source scanner of a document or image. Two types of approaches are commonly used:

A. Classical Machine Learning (Baseline)

- **Algorithms:** SVM, Random Forest, K-Nearest Neighbors (KNN)
- **Input:** Pre-extracted features such as:
 1. PRNU (Photo-Response Non-Uniformity)
 2. Texture descriptors
 3. Noise patterns
 4. Frequency domain features (FFT, wavelet transforms)
- **Workflow:**
 1. Feature extraction from raw scanned images
 2. Data normalization (e.g., `StandardScaler`)
 3. Train/test/validation split
 4. Model training and hyperparameter tuning (e.g., SVM kernel, C, gamma)
 5. Evaluation on validation/test sets (accuracy, F1-score, confusion matrix)

Output: Classifier capable of distinguishing scanners based on handcrafted features.

B. Deep Learning Approach (Milestone 3)

- **Algorithm:** Convolutional Neural Network (CNN)
- **Input:** Raw images (no manual feature extraction required)
- **Advantages:**
 - Learns hierarchical features automatically
 - Can capture complex noise patterns and textures that classical models may miss
- **Workflow:**

Step 1: Data Preparation

- Split raw images into train/validation/test sets
- Apply **data augmentation** to improve generalization:
 - Random rotation
 - Brightness/contrast adjustments
 - Horizontal/vertical flips
- Resize images to a consistent shape

Step 2: Model Building

- Use a CNN architecture:
 - Convolutional layers to extract local patterns
 - Pooling layers to reduce dimensionality
 - Fully connected layers for classification
- Example hyperparameters to tune:
 - Number of convolutional layers
 - Filter size and number of filters
 - Learning rate
 - Batch size
 - Dropout rate

Step 3: Model Training

- Loss function: **CrossEntropyLoss** for multi-class classification
- Optimizer: Adam or SGD
- Monitor **training curves**:
 - Training loss / validation loss
 - Training accuracy / validation accuracy

Step 4: Evaluation

- On **test set**, compute:
 - **Accuracy**: Overall performance

- **F1-score:** Balances precision and recall, useful for class imbalance
- **Confusion matrix:** Shows which scanners are misclassified

Step 5: Explainability

- Apply model interpretability tools:
 1. **Grad-CAM:**
 - Visualizes which regions of the scanned image influenced the CNN's decision
 - Highlights scanner-specific patterns or artifacts
 2. **SHAP (SHapley Additive exPlanations):**
 - Explains contributions of each pixel or region to the prediction
 - Can help forensic investigators understand scanner fingerprints

Output: A deep learning model capable of high-accuracy scanner classification with **explainable predictions**.

2. Milestone 3 Timeline Mapping

Week	Task	Details
Week 5	Build & Train CNN	- Load raw images - Apply augmentation - Train CNN and track training/validation curves
Week 6	Evaluate & Explain	- Compute accuracy, F1-score, confusion matrix - Apply Grad-CAM / SHAP to visualize scanner-specific patterns

3. Connection to Baseline Model

- Your **baseline SVM** serves as a **performance reference**.
- CNN is expected to **outperform SVM** because it learns **deep features automatically**, capturing subtle scanner-specific noise patterns.
- Validation/test metrics can be compared to show improvement from handcrafted features to deep learning features.